Appendix A: The Rebuild of a Spinning Black Hole

Companion to *Scalar-Tensor Hydrodynamics* (Version 3.15)

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Purpose and scope. Section 12.3 traces the continuous rebuild of a bound matter configuration — the Earth — through steps 0 to 12, showing how its 4×10^{51} confined patterns, whose builds borrow binding energy from the surrounding Ocean, slow local clocks by half their capacity share, tilt the surroundings through their spin, and send their record outward at the carrying speed c_s. This appendix applies the same generative accounting to the extreme regime of the framework: a spinning black hole, illustrated using Sagittarius A* at the center of the Milky Way. It unites the black-hole mechanics of Sections 8.1 and 8.2, the strong-field geometry and frame-dragging rules of Sections 11.7–11.9, and the rebuild law of Section 12 into a single continuous account.

A.1 The rebuild of Sagittarius A*

Sagittarius A*, seen from inside the Ocean, is not a gathered object and holds no confined patterns of its own. It is a vast, self-sustaining circulation of the Ocean's energy — about 4.3×10^6 solar masses of load — whose outer point of no return spans roughly 12.7×10^6 km in areal radius in the non-spinning limit, turning in place, sitting in the Milky Way's central eddy, and traveling with the galaxy at about 600 km s^{-1} relative to the frame in which the microwave background is uniform. Its rebuild, described at one moment (the rebuild is continuous; the order below is an order of description, not of ticks), runs as follows.

0. **What keeps it running.** The Ocean's flow powers a dynamo; the dynamo keeps the plexus running, the Ocean-wide waves that carry the re-made state; the flow's left twist is the twist every build carries. Across the black hole and its point of no return, the Ocean stays one connected state. Seen from afar, its phase turns at one rate everywhere, $\theta = \mu_\infty[t + F(r)]$, keeping in step near the edge not by turning infinitely fast and not by breaking into quantized vortex lines, but by falling freely inward, as every pattern does, across the point of no return; its current is made by the build at every place, so nothing is carried in and nothing piles up.

1. **Reclaiming and stripping.** Every part of the circulation just made is taken down, with a right twist. For any confined pattern that has fallen across the point of no return, this reclaiming is also its stripping. As the load rises and tides steepen across an infalling body, the pattern's form, its kind, its version, its lepton or baryon count, and its circulation sense (whether it entered as a build of matter or a reclaiming of antimatter) are unmade without annihilation. The black hole is the one setting in the Ocean where the matter/antimatter distinction is undone without a build meeting its partner.

2. **Handed forward.** What the reclaiming takes down supplies the build. Only the conserved carried quantities that couple to the Ocean's long-range state — total energy (mass M), net circulation (spin J), and electric charge (Q, whose long-range field in the Ocean must be the plexus that carries light) — are handed forward to the outside record, as the no-hair theorem requires. The fine-grained state is never destroyed; it is handed forward into the interior flow while the fallen patterns are scrambled.

3. **Building.** Each part of the circulation is built, with a left twist, on the plexus. Outside the core, no confined patterns belong to the black hole itself; what is built at each place is the re-made state — its ten numbers of energy (one), motion (three), and pushes (six) — turning in place.

4. **The build meets its own reclaiming.** Throughout the smooth circulation, the build and its reclaiming match, so the antimatter side stays hidden. Right at the point of no return, however, the steep gradient of the Ocean's state — which brings outward travel to zero — makes the clocks of the places there and the far clock part exponentially; plexus waves built fresh in step with their own place's clock are out of step with the far clock, and the mismatch leaves as outgoing ripples with partners inside, the energy drawn from the hole (Section 8.1). What leaves is energy, not what fell in: any localized patterns among the ripples are made afresh by the Ocean, as pairs. That is Hawking radiation: emerging from a region comparable in size to the hole (with a typical wavelength about eighty times the horizon radius), cold (about 1.4×10^{-14} K for Sagittarius A*, and reduced further by spin), shedding angular momentum faster than energy, and returning the hole's energy and state to the Ocean over immense times.

5. **The motion is part of the build.** Outside the point of no return, the net circulation of the Ocean within the hole is handed forward in each rebuild as a continuous sideways tilt s of the re-made state. Inside the ergosphere, that sideways tilt exceeds the local carrying speed c_s, so every rebuild carries whatever sits there sideways with the spin; no pattern can be rebuilt at rest relative to distant regions, though outward travel and escape remain possible. Past the outer point of no return, the direction in which time is made points inward: every rebuild carries every pattern toward the center.

6. **The scoop and rotational extraction.** The builds of a planet or star borrow only a small fraction of its mass from the surrounding Ocean as binding energy (about four parts in 10^{10} for the Earth). At the black hole, the scoop is total: at the point of no return, what the builds hold on loan equals the entire rest energy of whatever reaches it. Nothing inside can be rebuilt farther out unless that is paid back. Outside the point of no return but inside the ergosphere, however, a wave or pattern can be split so that one part falls in carrying tilt against the hole's spin, paying back part of what the hole's builds hold on loan for its spin and allowing the escaping part to leave with more energy than it brought in. That is rotational superradiance and, when threaded by magnetic fields of the accretion flow, the Blandford–Znajek beaming of jets along the spin axis [1].

7. **Slower where there is more to rebuild.** The rebuild uses a share of the Ocean's capacity for rebuilding ($c_0^4/2G$). For a non-spinning hole in isotropic coordinates, the clock rate seen from afar is

$$U = \frac{1 - GM/(2rc_0^2)}{1 + GM/(2rc_0^2)} = \sqrt{1 - \text{share}},$$

while $\psi^2 = \left(1 + GM/(2rc_0^2)\right)^2$ of the Ocean's own rulers fit in a unit step of r. At the photon orbit (areal radius $3GM/c_0^2$), two-thirds of the capacity is in use. At the point of no return (isotropic radius $GM/2c_0^2$, areal radius $2GM/c_0^2$), the share reaches one: seen from afar, a clock's pace reaches zero and no more time is made there, which is why the last image of a falling traveler reddens, dims, and never crosses. On the spot, a falling traveler's own clock and its rebuild are the same making of time, so the traveler crosses intact (experiencing a head-to-foot tidal difference of only about $10^{-4}\,g$ at the edge of Sagittarius A*) and notices nothing at the horizon.

8. **The surroundings are rebuilt with everything the hole carries.** Its energy sets the deep slowing of clocks and squeezing of rulers; its net circulation J sets the re-made tilt s that turns the surroundings continuously from the hole's own angular rate at the edge to the first-order gravitomagnetic field

$$s = \frac{2G}{c_0^2}\frac{J \times r}{r^3}$$

far away, with zero circulation on every loop of the scalar phase θ. Because the tilt carries both the clock share ($2uV$) and the ruler share ($2\gamma uV$ with $\gamma = 1$) of the hole's 600 km s^{-1} travel with the Milky Way, that bulk travel through the Ocean leaves no local turning trace on nearby orbits. When two black holes merge, the rapid rearrangement of their six push numbers ripples outward through the plexus at c_s as the two transverse stretch-and-squeeze polarizations of gravitational waves, ringing down as the newly merged circulation settles.

9. **The core turns the flow through.** Past the point of no return, forward in time points inward, and a falling pattern at Sagittarius A* reaches the center within $\pi GM/c_0^3 \approx 66$ s on its own clock, the most any pattern can have (letting go at the edge); one falling in from rest far away crosses moving and takes 28 s. The Ocean's stiffness ($\lambda > 0$) cannot turn this inward march around, because for any medium with $\varepsilon + p \geq 0$ inside a point of no return, contraction once under way only speeds up ($dH/dt = -4\pi G(\varepsilon + p)/c_0^2 \leq 0$). Instead, the Ocean's grip $\kappa(\phi)$, how strongly its rulers and clocks respond to what is in them, loosens under extreme compression, so that collapse slows, stops and reverses. The grip at a place is the part of the Ocean's capacity there not already drawn, read at the width of a dent (Section 11.9): with $\varepsilon/\varepsilon_c$ the share of a dent-width region, ε the energy density of what is in it, filled at ε_c, about 1.5 times the Planck energy density,

$$\kappa_{\text{eff}} = \kappa\left(1 - \frac{\varepsilon}{\varepsilon_c}\right), \qquad H^2 = \frac{8\pi G}{3c_0^2}\,\varepsilon\left(1 - \frac{\varepsilon}{\varepsilon_c}\right),$$

and the collapse stops at ε_c, at the center, 28 s after crossing the edge for a pattern falling in from rest far away. Nothing piles up or is crushed to a point. The rebuild stops the collapse short of a singularity and turns the inward flow around into an outward flow-through. Time is recycled with it: inward on the way in, outward again on the way out.

10. **The record spreads at the one speed.** Outside the point of no return, what the black hole has just been — its mass, its spin tilt, its charge, and its travel with the galaxy — spreads outward through the plexus at c_s, aimed at where the hole is now. From inside the point of no return, no immediate record can spread outward, because the inward flow of the rebuild outruns c_s; the reversed core flow must work its way back across the steep region near the edge by a random walk across the fine scale ℓ, taking of order $\tau \sim (GM/c_0^2)^{q+1}/(c_0\ell^q) \gtrsim 10^{39}$ years seen from afar for Sagittarius A*, with steps of a dent's width, before leaking out thermalized.

11. **Others rebuild themselves reading it.** The star S0-2, orbiting close to Sagittarius A*, reads the outside record across its body. Its near side is built more slowly than its far side, turning its travel inward; because the slowing composes in layers ($U \approx e^{-u}$ to second order, giving $\beta = 1$ and $\gamma = 1$), its pericenter advances by $(2 + 2\gamma - \beta)/3 = 1$ of General Relativity's rate, while the sideways tilt s turns its orbital plane, by the same term that turns a gyroscope's axis at $\frac{1}{2}\nabla \times s$. Light grazing 1.5 horizon radii is rebuilt so much more slowly on its inner flank that its path refracts into a closed orbit, casting the shadow of radius $3\sqrt{3}\,GM/c_0^2$ observed from the Earth.

12. **The loan is held, and the circulation just built is the one reclaimed next.** Unlike the Earth's builds, which return their small binding-energy loans to the Ocean as the Earth moves on at each moment, the black hole's builds hold their total scoop inside the point of no return from rebuild to rebuild, until early accretion and jet extraction outside, slow flow-through from the loosened core, Hawking radiation at the edge, or the global reclaiming at the end of the beat returns all of its energy and state to the Ocean.

A.2 Comparison of regimes: bound matter and a spinning black hole

Table 1 summarizes how the same rules of Section 12.7 operate across a bound planetary pattern (the Earth, Section 12.3) and a horizon-scale circulation (Sagittarius A*).

Rebuild Feature	The Earth (Section 12.3)	Spinning Black Hole (Sagittarius A*)
Composition	$\sim 4 \times 10^{51}$ confined patterns on gapped plexus waves	No confined patterns held; pure organized circulation of the re-made state
Phase & Flow (θ, u^μ)	The settled Ocean falls freely toward and through it, $11.2\,\mathrm{km\,s^{-1}}$ at the surface; no horizon	Smooth across the edge; the settled Ocean falls freely through it (Eq. (37)); zero phase circulation
Reclaiming & Sense	Matches build locally ($< 10^{-45}$ mismatch); radioactive β-decays release unmatched $\bar{\nu}_e$	Strips infalling patterns of kind, version, and matter/antimatter sense without annihilation
Capacity Share	1.4×10^{-9} at the surface; clocks slow by $\sim 7 \times 10^{-10}$	Reaches 1 at the point of no return ($2GM/c_0^2$); clock rate $U \to 0$ seen from afar
The Scoop (Loan)	Gravitational binding energy ($\sim 4 \times 10^{-10} M_\oplus$), returned as the Earth moves on	Total rest energy held on loan by the builds inside the horizon; the part held for its spin extractable in the ergosphere
Frame-Dragging Tilt	Re-made tilt s turns GP-B gyroscopes by -39.2 mas yr^{-1}	Tilt exceeds c_s in the ergosphere, enforcing co-rotation; falls off as $2G(J \times r)/(c_0^2 r^3)$
Interior Regularity	Hydrostatic pressure (360 GPa at center) supported by confined matter patterns	Stiffness cannot halt collapse; loosening of grip $\kappa(\phi)$ at ε_c turns collapse into flow-through
Return of State	Continuous as the pattern travels; geoneutrinos and heat escape freely	Early jets/accretion outside; Hawking radiation (10^{87} yr) and core random walk ($\gtrsim 10^{39}$ yr)

Table 1: Comparison of the continuous rebuild process for a bound matter body and a spinning black hole under the law of Section 12.7.

A.3 What this account settles and what remains open

Writing the spinning black hole in the rebuild's terms makes three structural features of the framework explicit:

1. **No vortex lines are required for Kerr-like frame dragging.** Because motion is part of the re-made state handed forward in Step 5 and Step 8, the sideways shift s is carried continuously by the plexus around the spinning hole while the scalar phase θ remains curl-free ($\nabla \times \nabla\theta = 0$) all the way across the outer point of no return. This removes the need for a high present-day chemical potential μ to pack quantized vortices together, preserving the stiffness measurement requires ($a \lesssim 10^{-23}$).

2. **Unitarity and the no-hair theorem coexist in Step 1 and Step 2.** Stripping is not the destruction of fine-grained state. It is the reclaiming of confined plexus patterns scrambled by the tide and the load, handing only the conserved quantities (M, J, Q) to the outside record while carrying the fine-grained state through the core turnaround (Step 9) and horizon pair-stretching (Step 4).

3. **Two open tasks from Section 14 govern the exact interior and near-horizon profile.** First, while the isotropic rules $\nabla \cdot (\psi^2 \nabla U) = 0$ and $\nabla^2 \psi = 0$ give the exact Schwarzschild geometry outside a non-spinning mass and Step 8 gives the Lense–Thirring tilt to first order in spin J, beyond first order in spin — where the Ocean's rulers cannot shrink isotropically in all three spatial directions — the plexus's frame law sets them by direction and gives the Kerr exterior, and with it the shadow distortion and merger ringdown spectrum (Section 11.8); the region within the inner edge remains. Second, deriving the grip's rule from the Ocean's law, and how the reversed flow crosses back, will fix the exponent q of the random-walk return delay in Step 10.

References

[1] R. D. Blandford and R. L. Znajek, *Electromagnetic extraction of energy from Kerr black holes*, Mon. Not. R. Astron. Soc. **179**, 433 (1977); doi:10.1093/mnras/179.3.433.

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